Firms and Market Structures

CFA Level I | LM01 | Concise revision sheet: the four market structures, with graphs

1. The four market structures at a glance

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Number of sellers	Many	Many	Few	Single firm
Barriers to entry / exit	Very low	Low	High	Very high
Product	Homogeneous (oranges, milk, wheat)	Close substitutes, differentiated (toothpaste)	Close substitutes or differentiated (airline on a route)	Unique, no good substitutes (utilities)
Non-price competition	None	Advertising, differentiation	Advertising, differentiation, features	Advertising
Pricing power	None: price taker	Some	Some to significant	Considerable
Demand curve faced by firm	Horizontal: P = MR = AR	Downward sloping	Downward sloping; pricing interdependent	Downward sloping (= market demand)
Supply function	Defined: MC above min AVC	Not well defined	Not well defined	Not well defined
Long-run economic profit	Zero	Zero	Possible (dominant firm's share tends to decline)	Positive (barriers protect it)

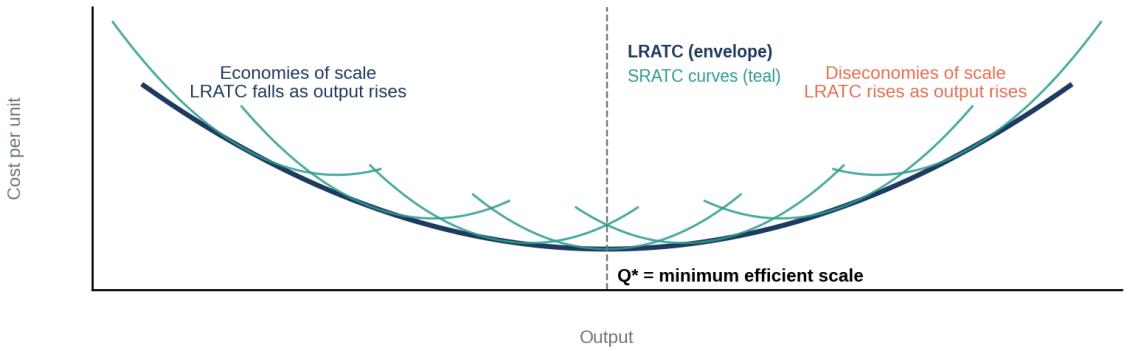
Five factors that determine market structure: (1) number and relative size of firms, (2) degree of product differentiation, (3) pricing power, (4) strength of barriers to entry and exit, (5) degree of non-price competition. Perfect competition and monopoly are the two extremes. Producers prefer monopoly/oligopoly (highest pricing power); consumers prefer perfect competition (lowest prices).

2. Foundations: profit maximisation, breakeven, shutdown, scale

- **Profit-maximising rule (all structures):** produce Q* where **MR = MC** and MC is rising (MR = MC on the falling part of MC is not a maximum). Price: for a price taker P = MR; for every other structure read P* off the **demand curve** at Q*.
- **Imperfect competition:** demand slopes down and MR lies below it (steeper). TR is maximised where MR = 0; MR > 0 where demand is elastic.
- **Breakeven:** TR = TC, or P = AR = ATC. Economic profit is zero (accounting profit may still be positive).

Situation	Short run	Long run
Price > ATC (TR > TC)	Economic profit: operate	Economic profit: operate (entry in competitive markets)
Price = ATC	Breakeven: operate	Breakeven: operate
AVC < Price < ATC (TVC < TR < TC)	Operate (covers variable cost, losses limited)	Exit
Price < AVC (TR < TVC)	Shut down	Exit

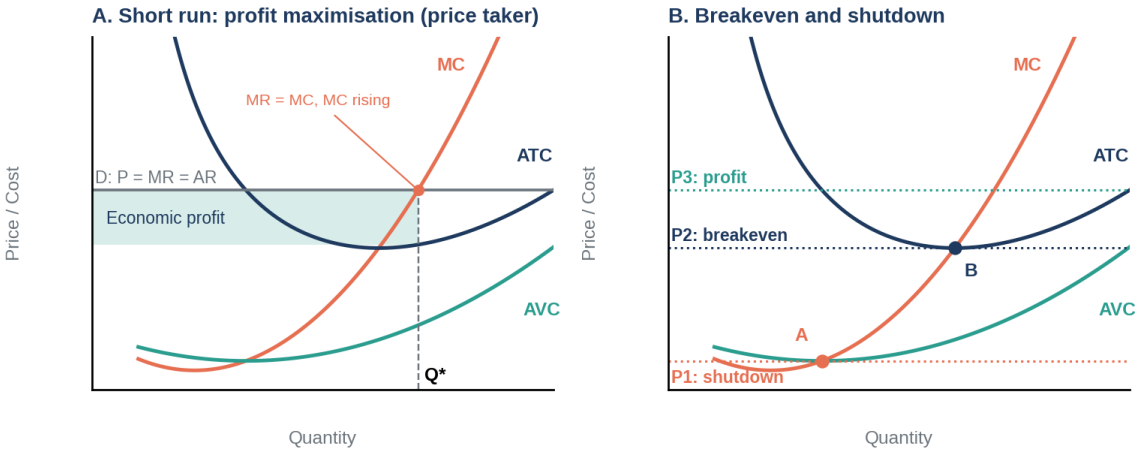
Long-run average total cost (LRATC) as envelope of SRATC curves



Economies of scale: LRATC falls as output rises (increasing returns, division of labour, better equipment, bulk buying, quality control). Diseconomies: LRATC rises (decreasing returns, management complexity, supply bottlenecks, higher labour costs). Q* is the minimum efficient scale (lowest LRATC).

3. Perfect competition

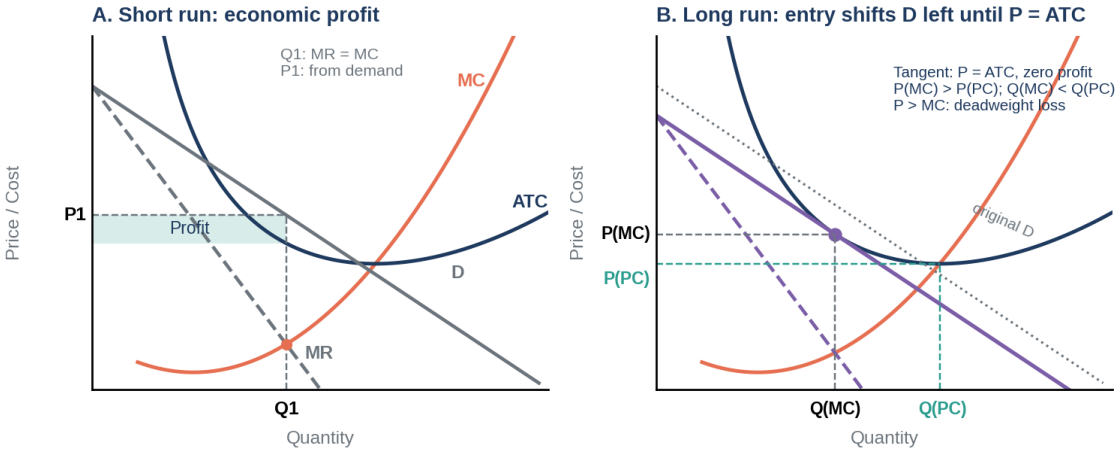
- **Features:** many buyers and sellers, identical product, no barriers, no pricing power. The firm is a **price taker**: horizontal demand, P = MR = AR, TR rises by exactly P for each extra unit.
- **Short-run supply** = the MC curve above minimum AVC. A higher market price shifts the demand/MR line up and raises Q*; a lower price reduces Q*.
- **Long run:** economic profit attracts entry, which pushes the market price down until P = ATC (zero economic profit). Losses cause exit until the same point.
- **Efficiency:** P = MC, so total surplus is maximised and there is no deadweight loss. Price is lowest and quantity highest of all structures, but consumers get little variety.



Left: P is above ATC at Q (MR = MC), so the shaded rectangle is economic profit. Right: P3 gives profit; P2 (min ATC) is breakeven; P1 (min AVC) is the shutdown price. Between A and B the firm operates in the short run despite losses.*

4. Monopolistic competition

- **Features:** many sellers of close but **differentiated** substitutes (e.g. burgers, toothpaste); low entry costs; some pricing power; competition through advertising and product differentiation.
- **Demand:** downward sloping; MR below demand. **Output:** MR = MC; **price** from the demand curve. Short-run profit = (P1 - ATC1) x Q1 = TR - TC.
- **No well-defined supply curve:** the firm sets its own price, so MC alone does not show quantity supplied at each price.
- **Long run:** short-run profit invites entry. Each new firm shifts every existing firm's demand and MR curves to the left (lower market share) until **P = ATC**: zero economic profit, no more entry. Demand is tangent to ATC.



Left: short-run profit at Q1. Right: long-run equilibrium after entry. P(MC) > P(PC) and Q(MC) < Q(PC) compared with perfect competition.

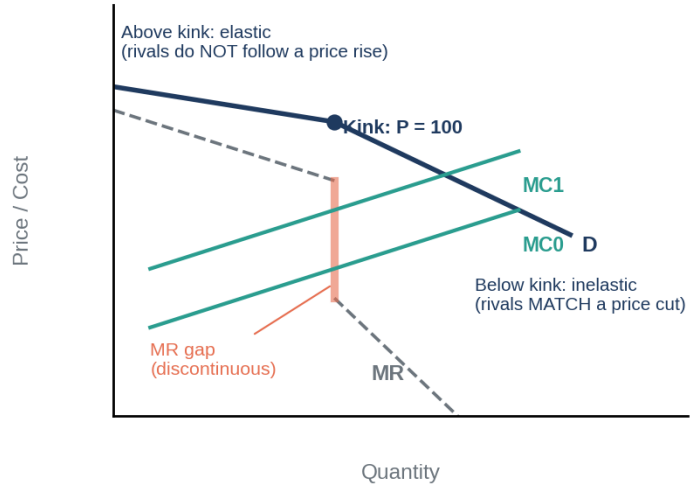
Versus perfect competition: both earn zero economic profit in the long run and both maximise at MR = MC. But perfect competition has P = MC (efficient, surplus maximised) while monopolistic competition has **P > MC** and a deadweight loss, with lower output and higher price, plus advertising costs; in return consumers get more variety.

5. Oligopoly

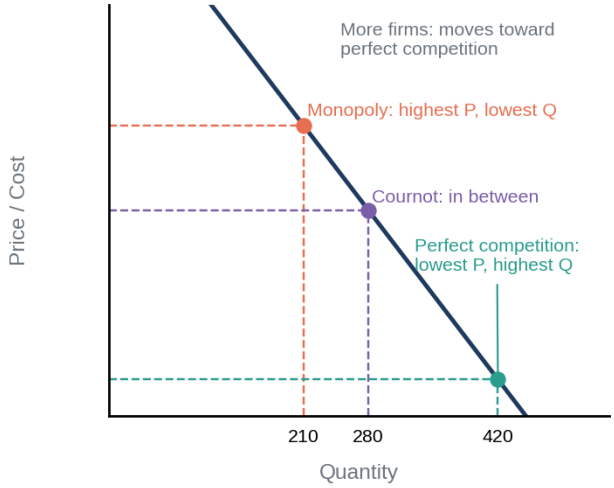
- **Features:** few large sellers; products are close substitutes (branded or not); high barriers; substantial pricing power. Pricing is **strategic and interdependent**: each firm must anticipate rivals' reactions, and there is a temptation to **collude** (a cartel if open).
- **No single optimal price/output model.** Supply function not well defined. Rule is still MR = MC, with price from demand, but the outcome depends on the strategic model used:

Model	Core idea	Price / output outcome
Kinked demand (pricing interdependence)	Rivals ignore a price rise but match a price cut, so demand is elastic above the kink and inelastic below. MR is discontinuous.	Price at the kink. Stays stable while MC moves within the MR gap. Explains price stability but not what the price should be.
Cournot	Firms choose output simultaneously, each assuming rivals' output stays fixed.	Price and quantity between monopoly and perfect competition. More firms moves the result toward perfect competition.
Nash equilibrium	Each firm does its best given rivals' strategies; no collusion. No firm gains by changing strategy alone.	No certainty of price/output; depends on the payoff matrix.
Stackelberg / dominant firm	One large low-cost leader (about 40% or more share) sets price and moves first; small followers are price takers.	Leader: MR(L) = MC(L), price from D(L). Followers supply the rest at the leader's price.

A. Kinked demand curve (price interdependence)



B. Cournot vs other structures



Left: the kink at P = 100 (Q = 5,000) creates a gap in MR; MC0 and MC1 both cut MR inside the gap, so price does not change. Right: Cournot lies between monopoly and perfect competition (schematic numbers from the notes).

Nash equilibrium example (profits shown as W = WesCo, R = RifCo)

	RifCo: Low price	RifCo: High price
WesCo: Low price	W 50, R 70	W 80, R 0
WesCo: High price	**W 300, R 350** Nash equilibrium	W 500, R 300 (total 800)

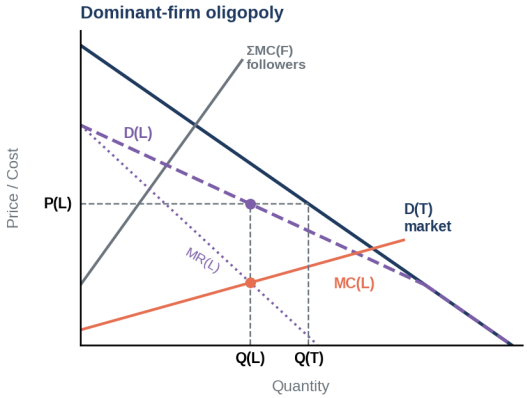
- If WesCo is High, RifCo does better by cutting to Low (350 vs 300). If RifCo is Low, WesCo does better staying High (300 vs 50). Neither wants to move: **Nash equilibrium**.
- Both High (total 800) beats Nash (650), but each side is tempted to cheat. If they **collude** and split 800 equally, each earns 400, above 300 and 350.

Chances of successful collusion

- Few sellers, similar size
- Homogeneous products
- Similar cost structures
- Small, frequent orders
- Severe threat of retaliation for cheating
- Little external competition (collusion raises profits and attracts entrants)

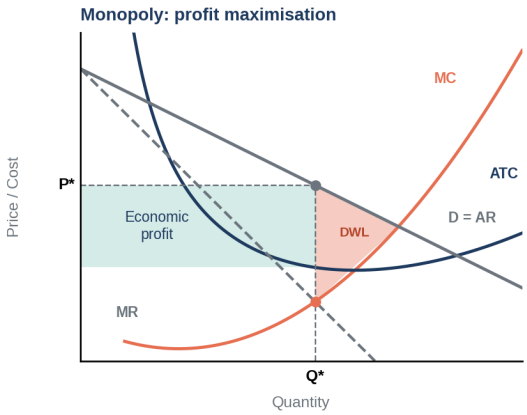
Dominant-firm oligopoly (the supply case in the curriculum)

- Leader: lowest cost, at least 40% share, greater capacity, **price maker**. Followers: small price takers.
- D(L) = market demand D(T) minus followers' supply ΣMC(F). Leader picks Q(L) where MR(L) = MC(L) and charges P(L) from D(L).
- Followers supply Q(T) - Q(L) at P(L). They will not undercut: a price war with a low-cost leader is a losing fight.
- D(L) and D(T) converge as price falls: followers exit rather than sell below cost, so the leader's share rises.
- **Stackelberg** is sequential (leader chooses output first, follower second); Cournot is simultaneous.



6. Monopoly

- **Features:** a single seller of a unique product with no good substitutes; very high barriers to entry (legal, control of resources, scale). Utilities such as electricity and water are typical, often controlled or regulated by government.
- **Demand** is the market demand: downward sloping, MR below it.
- **Output:** MR = MC gives Q*. **Price:** read P* from the demand curve at Q* (not from MC). Same rule as other structures, but P* > MR = MC.
- **Profit:** P* > ATC, so economic profit is positive and, because entry is blocked, it **persists in the long run**.
- **Efficiency:** highest price and lowest quantity of all structures; P > MC creates a **deadweight loss** (the shaded triangle).



7. Identifying market structure: concentration measures

- **Why it matters:** analysts favour industries with pricing power, and regulators use competition (antitrust) law to block mergers that reduce competition.
- **Econometric approach:** regression to estimate demand and supply elasticity; inelastic demand suggests market power. Theoretically appealing but data are hard to obtain.

Measure	Calculation	Range / reading	Example
N-firm concentration ratio	Sum of the market shares of the largest N firms	About 0 for perfect competition, 100% for monopoly	Top five firms with 25%, 15%, 10%, 10%, 10% give 70%
Herfindahl-Hirschman Index (HHI)	Sum of squared market shares of the N largest firms (decimals)	0 to 1: near 0 competitive, 1 = monopoly	Shares 50%, 20%, 10%, 20% give 0.25 + 0.04 + 0.01 + 0.04 = 0.34

- **N-firm ratio:** easy to compute with readily available data, but not affected by mergers among the top firms and does not quantify market power.
- **HHI:** simple and commonly used by regulators; squaring gives large firms more weight. **Both** ignore barriers to entry and elasticity of demand.

Exam checklist

- Rule for every structure: **MR = MC**. Only the price-taker has P = MR; everyone else prices off the demand curve.
- Shutdown if P < AVC (short run); exit in the long run if P < ATC. Breakeven at P = ATC = MC (minimum ATC for a competitive firm).
- Long-run economic profit: zero in perfect and monopolistic competition; possible in oligopoly; persistent in monopoly. Only perfect competition has P = MC.
- Oligopoly models: kinked demand (price stability), Cournot (simultaneous output), Nash (no incentive to deviate), Stackelberg/dominant firm (leader moves first). Collusion is easier with few, similar firms and credible retaliation.